

1. Start by entering your V_{IN} , V_{OUT} and I_{OUT} requirements.
2. Optimize your design for key parameters like efficiency, footprint or cost using the optimizer dial and compare this design with other possible solutions from Texas Instruments.
3. WEBENCH Power Designer provides you with a customized schematic along with a list of materials with real time pricing and component availability.
4. In most cases, you will also be able to:
 - Run electrical simulations to see important waveforms and circuit performance,
 - Run thermal simulations to understand the thermal performance of your board,
 - Export your customized schematic and layout into popular CAD formats,
 - Print PDF reports for the design, and share your design with colleagues.
5. Get more information about WEBENCH tools at www.ti.com/webench.

8.2.2.2 Inductor Selection

The inductor selection is affected by several parameters such as inductor ripple current, output voltage ripple, transition point into Power Save Mode, and efficiency. See [Table 8-2](#) for typical inductors.

For high efficiencies, the inductor must have a low DC resistance to minimize conduction losses. Especially at high-switching frequencies, the core material has a high impact on efficiency. When using small chip inductors, the efficiency is reduced mainly due to higher inductor core losses. This needs to be considered when selecting the appropriate inductor. The inductor value determines the inductor ripple current. The larger the inductor value, the smaller the inductor ripple current and the lower the core and conduction losses of the converter. Conversely, larger inductor values cause a slower load transient response. To avoid saturation of the inductor, the peak current for the inductor in steady state operation is calculated using [Equation 5](#). Only the equation which defines the switch current in boost mode is shown, because this provides the highest value of current and represents the critical current value for selecting the right inductor.

$$\text{Duty Cycle Boost} \quad D = \frac{V_{OUT} - V_{IN}}{V_{OUT}} \quad (4)$$

$$I_{PEAK} = \frac{I_{out}}{\eta \times (1 - D)} + \frac{V_{in} \times D}{2 \times f \times L} \quad (5)$$

where:

- D = Duty Cycle in Boost mode
- f = Converter switching frequency
- L = Inductor value
- η = Estimated converter efficiency (use the number from the efficiency curves or 0.9 as an assumption)

Note

The calculation must be done for the minimum input voltage in boost mode.

Calculating the maximum inductor current using the actual operating conditions gives the minimum saturation current of the inductor needed. It is recommended to choose an inductor with a saturation current 20% higher than the value calculated using [Equation 5](#). Possible inductors are listed in [Table 8-2](#).

Table 8-2. List of Recommended Inductors

INDUCTOR VALUE [μ H] ⁽¹⁾	SATURATION CURRENT [A]	DCR [m Ω]	PART NUMBER	MANUFACTURER	SIZE (LxWxH mm)
2.2	3.5	21	XFL4020-222ME	Coilcraft	4 x 4 x 2
2.2	1.7	72	SRN3015TA-2R2M	Bourns	3 x 3 x 1.5
2.2	3.1	97	DFE252010F-2R2M	Murata	2.5 x 2 x 1
2.2	2.4	116	DFE201612E-2R2M	Murata	2.0 x 1.6 x 1.2

Table 8-2. List of Recommended Inductors (continued)

INDUCTOR VALUE [μ H] ⁽¹⁾	SATURATION CURRENT [A]	DCR [m Ω]	PART NUMBER	MANUFACTURER	SIZE (LxWxH mm)
2.2	2.0	190	DFE201210U-2R2M	Murata	2.0 x 1.2 x 1.0

(1) See the [Third-party Products Disclaimer](#).

8.2.2.3 Output Capacitor Selection

For the output capacitor, use of small ceramic capacitors placed as close as possible to the VOUT and GND pins of the IC is recommended. The recommended nominal output capacitor value is a single 22 μ F. If, for any reason, the application requires the use of large capacitors which cannot be placed close to the IC, use a smaller ceramic capacitor in parallel to the large capacitor. The small capacitor must be placed as close as possible to the VOUT and GND pins of the IC.

It is important that the effective capacitance is given according to the recommended value in [Section 6.3](#). In general, consider DC bias effects resulting in less effective capacitance. The choice of the output capacitance is mainly a tradeoff between size and transient behavior as higher capacitance reduces transient response overshoot and undershoot and increases transient response time. Possible output capacitors are listed in [Table 8-3](#).

There is no upper limit for the output capacitance value.

At light load currents the output voltage ripple is dependent on the output capacitor value. Larger output capacitors reduce the output voltage ripple. The leakage current of the output capacitor adds to the overall quiescent current.

Table 8-3. List of Recommended Capacitors

CAPACITOR VALUE [μ F] ⁽¹⁾	VOLTAGE RATING [V]	PART NUMBER	MANUFACTURER	SIZE (METRIC)
22	6.3	GRM187R60J226ME15	Murata	0603 (1608)
22	6.3	GRM219R60J476ME44	Murata	0805 (3210)
47	6.3	GRM188R60J476ME15	Murata	0603 (1608)

(1) See [Third-party Products Disclaimer](#).

8.2.2.4 Input Capacitor Selection

A 10- μ F input capacitor is recommended to improve line transient behavior of the regulator and EMI behavior of the total power supply circuit. An X5R or X7R ceramic capacitor placed as close as possible to the VIN and GND pins of the IC is recommended. This capacitance can be increased without limit. If the input supply is located more than a few inches from the TPS63900 converter additional bulk capacitance can be required in addition to the ceramic bypass capacitors. An electrolytic or tantalum capacitor with a value of 47 μ F is a typical choice.

When operating from a high impedance source, a larger input buffer capacitor is recommended to avoid voltage drops during start-up and load transients.

The input capacitor can be increased without any limit for better input voltage filtering. The leakage current of the input capacitor adds to the overall quiescent current.

Table 8-4. List of Recommended Capacitors

CAPACITOR VALUE [μ F] ⁽¹⁾	VOLTAGE RATING [V]	PART NUMBER	MANUFACTURER	SIZE (METRIC)
10	6.3	GRM188R60J106ME47	Murata	0603 (1608)
10	10	GRM188R61A106ME69	Murata	0603 (1608)
22	6.3	GRM187R60J226ME15	Murata	0603 (1608)

(1) See [Third-party Products Disclaimer](#).

8.2.2.5 Setting The Output Voltage

The output voltage is set with CFGx pins (see [Section 7.3.6](#)).